

University of Tartu
Faculty of Science and Technology
Institute of Ecology and Earth Sciences
Department of Geography

Project

Suitability analysis of car charging stations in Estonia

Author: Mihhail Batura

Supervisors: PhD Alexander Knoch

PhD Anto Aasa

PhD Holger Virro

MSc Pamela Maricela Guaman Pintado

Tartu 2025

Introduction

Electric cars have become increasingly popular in recent years. However, to use such machines fully, they need to be constantly charged. Such complex mechanisms as cars need to be charged very often and expensive charging stations are required. Therefore, it was useful for users to know how accessible charging stations are for their cars in their country.

The aim of this project is to illustrate the availability of charging stations in Estonia. The suitability analysis was taken as a basic method. To analyze the distance to the nearest station, the entire territory of Estonia was selected. To analyze the travel time to the nearest station, the region of southeastern Estonia was selected.

Data

Four files were used as initial data: charging stations (points), roads (lines), rivers (lines), counties (polygons).

Charging stations - car charging points in Estonia, designed in OpenChargeMap, <https://openchargemap.org/site>

Counties - administrative division of Estonia, designed in Estonian Landboard, <https://geoportaal.maaamet.ee/est/ruumiandmed/haldus-ja-asustusjaotus-p119.html>

Roads - roads with asphalt surfaces of various classes, taken from Spatial data analysis course (LOOM.02.016), EST_roads.zip

Rivers - watercourses that act as barriers to movement, taken from Cartography course (LOOM.02.009), jõed.zip.

Aunap R, i.a., *Kiirabi kohalejõudmise prognooskaardi modelleerimine*, <https://moodle.ut.ee/course/view.php?id=445>

Methodology

At the first step, the distance to the nearest charging station was calculated in Estonia. According to the OpenChargeMap, there are 167 charging stations in Estonia. As a result, a distance matrix (raster file) was created to display the distance to charging stations within the territory of Estonia. Analysis was carried out in the Rstudio software, *Batura_fproj1.R*.

At the second and the third step, the travel time to the nearest station was calculated in minutes. The sample included seven counties of Estonia: Järva County, Jõgeva County, Tartu County, Põlva County, Võru County, Valga County, and Viljandi County. At the second step, charging stations, roads were masked by selected counties. There are 55 charging stations within these counties. The islands were excluded from the sample because they complicate the cost distance calculation. To calculate travel time, roads and rivers were needed as guides and barriers. The roads were divided into 5 classes, presented in table 1. The class “0” is an empty field for rasterizing the road and river layer (table 2).

Table 1. Average estimated speed on different types of roads.

level	road type	speed (km/h)
0	field	0,1
1	main road	90
2	basic road	70
3	connectivity road	50
4	secondary road	50
5	street	30

Table 2. A speed multiplier at a certain section of the barrier.

level	type	speed (km/h)
0	field	1
1	river	0.1

Results

During the analysis, two maps were created: a distance matrix for the whole of Estonia and travel time to charging stations in south-eastern Estonia. The distance matrix is shown in Figure 2. The travel time map to the nearest charging station is shown in Figure 3.

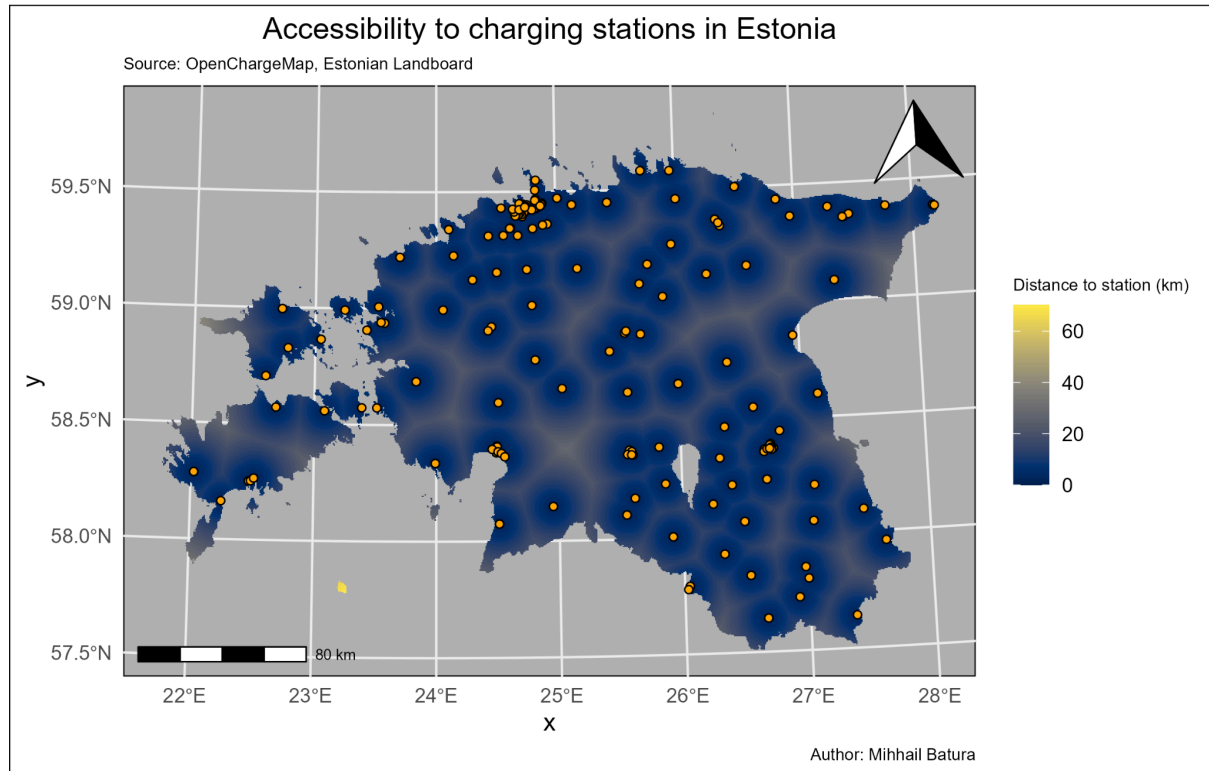


Figure 2. Distance matrix map with a distance to the nearest charging station.

In the picture you can see that the island of Ruhnu is noticeably distant from the rest of the part, as a result of which its distance value in relation to others stands out noticeably. It is also noticeable that most of the stations are located within the cities or close to them.

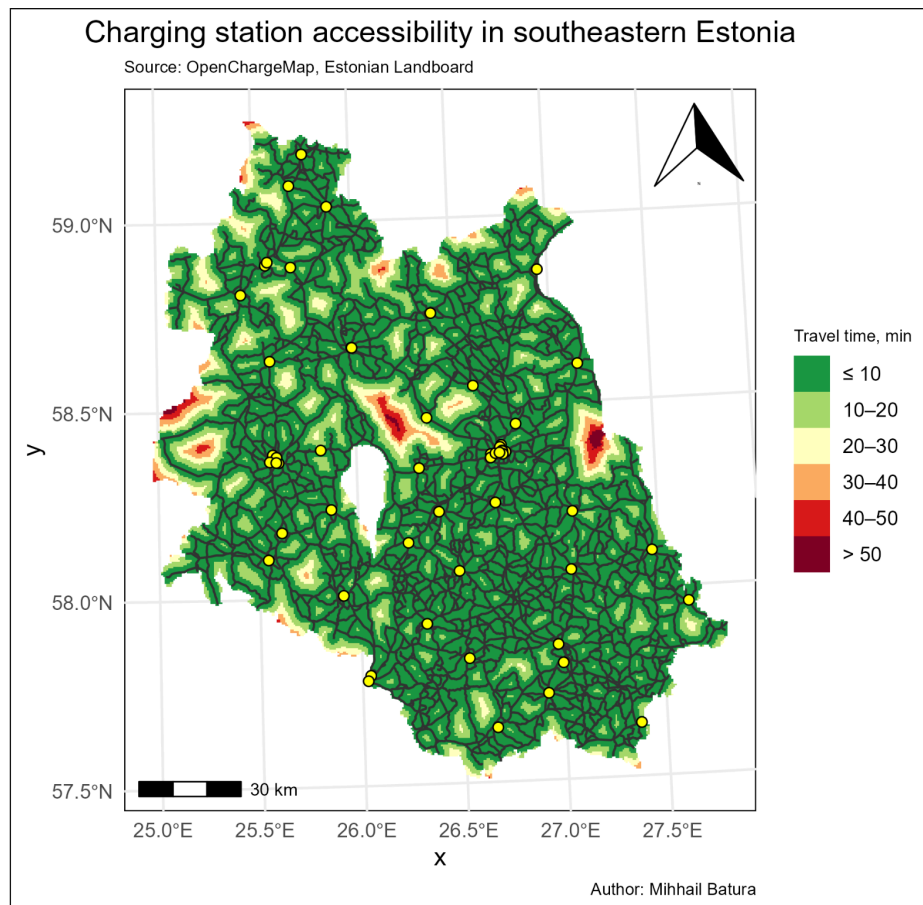


Figure 3. Travel time map to the nearest charging station in southeastern Estonia.

Figure 3 shows that the average travel time to the nearest charging station is no more than 10 minutes, given ideal conditions and the average estimated speed on a given section of road. Difficulties with accessibility of charging stations might arise if the car gets stuck on country roads in the middle of forests or swamps where there are no paved roads.